

The Hierarchy Law of Mathematical Languages

Eight clauses on five closure operators over a finite index — fully derived, with provenance.

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§0 · The law

Five formal languages close the same finite data, and their disagreement is lawful.

A finite table of observations — records over d attributes — does not speak for itself. Ask what *else* must hold given what was seen, and the answer depends on the language the question is put in. Order answers with a staircase, algebra with a sublattice hull, geometry with a pairwise convex hull, information with a join-closure, statistics with a pairwise-marginal closure. Each returns a superset of the data. They need not return the same one.

Their relations are not free. **Of the twenty containments among the five, exactly seven hold in every finite index** — four relations, once **order** and **algebra** are recognised as the single operator they are. Each of the seven is proved, with the one imported step named where it is used: Clause B's lift to $d \geq 3$ is cited rather than reproved here (§4.3, §9), and Lemma 5 proves it outright at $d = 2$. **The other thirteen all vary from one index to the next** — six of them refuted as universal claims by an explicit minimal witness, the rest by measurement alone. Two of the five languages are one operator (Clause B). One is contained in another as a theorem rather than the tendency it had been recorded as (Clause C). Four of the five require the order of the index and the fifth does not (Clause F) — and those four require *different* things, each language being invariant under its own group of relabellings, the four groups strictly nested in an order that is not the containment hierarchy (F.2). And the total ranking in which all five were supposed to nest is **not** part of the law — it is a property of whichever index it was measured on, and fourteen distinct rankings appear across four hundred (Clause E).

It is a law of applied mathematics, and it does not stop there. Carrying one body of data in several formal languages at once is the ordinary condition of an applied science. Physics is the extreme case: a single system held simultaneously in geometry, algebra, information and statistics, with relations between those descriptions relied on daily and never separated into the necessary and the incidental. Clauses E and H perform that separation. What is proved below is proved for these five operators over finite products of chains, and the distinction it draws is not peculiar to them.

What is not settled here. Which operators belong to the family at all: §8b and §8c weigh two further candidates and admit neither. And anything off a finite product of chains — §8's N2 is Clause B failing there, in one direction only.

Five status words are used and never merged. **The distinction between the third and fourth is the one most easily lost, and this document has lost it before.**

STATUS	MEANING
PROVED	a proof is written out below and every step is justified
MACHINE-CHECKED	Z3 returned unsat on the negation of an obligation whose variables range over <i>every</i> subset of a named finite box. §10 says which objects carry it, over which boxes, and which fragment of each was encoded
EXHAUSTIVE	a decision procedure visited <i>every</i> case in a stated finite family
E	
SAMPLED	a seeded pseudorandom sweep of a stated size. Not exhaustive. A figure like "300/300" is a sample result, and §10 marks these separately
CITED	taken from the literature; marked READ or UNREAD

Two further words appear as markers rather than statuses: **REFUTATION** (§7, a claim disproved) and **CANDIDATE** (§8b, §8c, operators outside the five, filed as evidence).

The ceiling that is real: machine-checking here decides claims over a *named finite box*, so nothing above $d = 3$ and nothing over an unbounded box carries that status. What is machine-checked is named in §10 and nowhere else.

§1 • Notation and definitions

Throughout, $d \geq 2$ is finite and X is a finite non-empty set of d -tuples.

D1 (observed alphabet). $A_i := \pi_i(X)$, the set of values X realises at coordinate i . Each A_i carries a total order, so each is a **finite chain**.

D2 (box). $\text{Box}(X) := A_1 \times \dots \times A_d$. As a product of chains it is a lattice under coordinatewise $\wedge = \min$ and $\vee = \max$. Any subset of a chain is a sublattice of it, so $\text{Box}(X)$ is closed under both.

D3 (generated sublattice). $\langle X \rangle :=$ the smallest subset of $\text{Box}(X)$ containing X and closed under $\wedge$ and $\vee$. Since $\wedge$ and $\vee$ act coordinatewise, no new coordinate value is ever created, so the closure stays inside $\text{Box}(X)$. This is what `op_algebra` computes.

D4 (boundary function). For $i \neq j$ and $a \in A_j$,

$$\varphi_{ij}(a) := \max \{ y_i : y \in X, y_j \leq a \}, \text{ undefined if that set is empty.}$$

D5 (staircase). $R(X) := \{ x \in \text{Box}(X) : x_i \leq \varphi_{ij}(x_j) \text{ for all } i \neq j \}$. This is what `op_order` computes.

D6 (join-closure). $J(X) :=$ the smallest superset of X closed under $\vee$ alone.

D7 (geometry). $G(X) := \{ x \in \text{Box}(X) : (x_i, x_j) \in \text{conv}(\pi_{ij} X) \text{ for all } i < j \}$.

D8 (statistics, order 2). $\text{St}(X) := \{ x \in \text{Box}(X) : \pi_{ij}(x) \in \pi_{ij}(X) \text{ for all } i < j \}$.

D9 (pair-definable). An operator L is *pair-definable* if there exist $C_{ij} \subseteq A_i \times A_j$ with $L(X) = \{ x \in \text{Box} : (x_i, x_j) \in C_{ij} \text{ for all } i < j \}$.

D10 (k-determined). $S \subseteq \text{Box}$ is *k-determined* if $S = \{ x \in \text{Box} : \text{every } k\text{-fold projection of } x \text{ is a } k\text{-fold projection of some element of } S \}$.

Two ambient regimes, distinguished because conflating them caused an error earlier in this work:

- **own-box**: each set is read in its own $\text{Box}(\cdot)$;
- **fixed-box**: $X \subseteq Y$ are both read in one declared Box (which may exceed either observed box).

Clause A is proved in **both**. Clause B holds in the **own-box** regime and can fail in the other (§8, N1).

§2 · Preliminaries

Lemma 1 (ϕ is total and isotone in its argument). For $a \in A_j$ there is $y \in X$ with $y_j = a$ (D1), so $\{y \in X : y_j \leq a\} \neq \emptyset$ and $\phi_{ij}(a)$ is defined. If $a \leq a'$ the candidate set for a' contains that for a , so $\phi_{ij}(a) \leq \phi_{ij}(a')$. ■ **PROVED**.

D1 is used in exactly two places, and this note previously claimed one. Here, to make ϕ total; and in Lemma 5, where two of the four witnesses are produced by it ($a \in A_1 = \pi_1(X)$). Nowhere else.

Lemma 2 (ϕ is monotone in X). For $X \subseteq Y$ in a fixed box and any $i \neq j$, $a: \{y_i : y \in X, y_j \leq a\} \subseteq \{y_i : y \in Y, y_j \leq a\}$, so $\phi^X_{ij}(a) \leq \phi^Y_{ij}(a)$ whenever the left side is defined; and where ϕ^X is undefined the staircase rejects outright. ■ **PROVED**, and **SAMPLED**: 0 failures over 400 nested pairs.

Lemma 3 (the staircase does not move its own boundary). $\phi^{\{R(X)\}}_{ij} = \phi^X_{ij}$ on $\text{Box}(X)$. *Proof.* $X \subseteq R(X)$ (Lemma 4a) gives $\geq$ by Lemma 2. For $\leq$: let $y \in R(X)$ with $y_j \leq a$. Then $y_i \leq \phi^X_{ij}(y_j) \leq \phi^X_{ij}(a)$ by Lemma 1. So every candidate contributed by $R(X)$ is bounded by $\phi^X_{ij}(a)$, hence the max is too. ■ **PROVED**, 0 failures over 400 cases.

§3 · Clause A — every admitted language is a closure operator

A. Each of `order`, `algebra`, `geometry`, `information`, `statistics` is **extensive**, **monotone** and **idempotent** on subsets of a fixed box.

This answers a correction made during the work: the axioms had been *measured* 300/300 and never derived. Each is now derived.

A.1 Extensive.

- `order`: for $x \in X$ and any $i \neq j$, x itself witnesses $x_j \leq x_j$, so $\phi_{ij}(x_j) \geq x_i$. ■
- `algebra`, `information`: immediate from D3, D6. ■
- `geometry`: $\pi_{ij}(x) \in \pi_{ij} X \subseteq \text{conv}(\pi_{ij} X)$. ■
- `statistics`: $\pi_{ij}(x) \in \pi_{ij} X$. ■

A.2 Monotone ($X \subseteq Y$, fixed box).

- `order`: by Lemma 2 every constraint is weaker for Y , so $R(X) \subseteq R(Y)$. ■
- `algebra`: $\langle Y \rangle$ is a sublattice containing X , hence contains $\langle X \rangle$. ■
- `information`: $J(Y)$ is join-closed and contains X , hence contains $J(X)$. ■
- `geometry`: `conv` is monotone. ■

- **statistics**: $\pi_{ij} X \subseteq \pi_{ij} Y$. ■

In the **own-box** regime, **order** and **algebra** inherit monotonicity from $\langle \cdot \rangle$ via Clause B instead — **SAMPLED** at 4,000 nested pairs, 0 failures (own-box regime).

A.3 Idempotent.

- **order**: by Lemma 3 the boundary functions of $R(X)$ equal those of X , so $R(R(X)) = R(X)$. ■
- **algebra**, **information**: a closure is closed. ■
- **geometry**: $\pi_{ij} X \subseteq \pi_{ij} G(X) \subseteq \text{conv}(\pi_{ij} X)$, and **conv** of a set between S and $\text{conv}(S)$ is $\text{conv}(S)$; so the hulls are unchanged and $G(G(X)) = G(X)$. ■
- **statistics**: $\pi_{ij} \text{St}(X) \subseteq \pi_{ij} X$ by definition, and $\supseteq$ because each witnessing element of X

lies in $\text{St}(X)$; so the marginals are unchanged. ■

Status: PROVED, all five, both regimes. Cross-checked **SAMPLED** at 300/300 worlds.

§4 • Clause B — **order** and **algebra** are one operator

B. $R(X) = \langle X \rangle$. Equivalently $\text{op_order} = \text{op_algebra}$, and both compute the sublattice hull.

4.1 The easy inclusion

Lemma 4. $R(X)$ is a sublattice of $\text{Box}(X)$ containing X . (a) Extensivity is A.1. (b) Let $p, q \in R(X)$ and fix $i \neq j$.

- **Join**. WLOG $p_i \geq q_i$. Then $(pvq)_i = p_i \leq \phi_{ij}(p_j) \leq \phi_{ij}((pvq)_j)$ by Lemma 1.
- **Meet**. WLOG $p_j \leq q_j$, so $(p \wedge q)_j = p_j$. Then $(p \wedge q)_i \leq p_i \leq \phi_{ij}(p_j) = \phi_{ij}((p \wedge q)_j)$. ■

Hence $\langle X \rangle \subseteq R(X)$, using nothing but Lemma 1. The content is the reverse.

4.2 The two-dimensional case

Lemma 5. For $d = 2$, $R(X) \subseteq \langle X \rangle$. *Proof.* Let $(a, b) \in R(X)$. Four witnesses exist in X :

WITNESS	EXISTS BECAUSE
y with $y_A = a$	$a \in A_1 = \pi_1(X)$ (D1)
z with $z_B = b$	$b \in A_2 = \pi_2(X)$ (D1)
u with $u_A \leq a$, $u_B \geq b$	$b \leq \phi_{BA}(a)$, and the max is attained
v with $v_A \geq a$, $v_B \leq b$	$a \leq \phi_{AB}(b)$, and the max is attained

Set $\alpha := y \vee u$ and $\gamma := z \vee v$. Then $\alpha = (\max(a, u_A), \max(y_B, u_B)) = (a, \beta)$ with $\beta \geq u_B \geq b$, and $\gamma = (\max(z_A, v_A), \max(b, v_B)) = (\alpha', b)$ with $\alpha' \geq v_A \geq a$. Therefore

$$\alpha \wedge \gamma = (\min(a, \alpha'), \min(\beta, b)) = (a, b),$$

and $\alpha, \gamma \in \langle X \rangle$, so $(a, b) \in \langle X \rangle$. ■ **PROVED**. EXHAUSTIVE: the construction was *built*, not inferred, on 54,392 cells of R , 0 failures.

4.3 The lift

Lemma 6. π_{ij} is a lattice homomorphism, so $\pi_{ij}(X) = \langle \pi_{ij} X \rangle$. Also φ_{ij} depends only on $\pi_{ij} X$, and $\pi_i(\pi_{ij} X) = A_i$, so D1 passes to the projection. With Lemma 5, $\langle \pi_{ij} X \rangle = R_{ij} := \{(s,t) : s \leq \varphi_{ij}(t), t \leq \varphi_{ji}(s)\}$. ■

Lemma 7 (the lift). $\langle X \rangle = \{ x \in \text{Box}(X) : (x_i, x_j) \in \pi_{ij}(X) \text{ for all } i < j \}$.

With Lemma 6 the right-hand side is exactly D5, giving **B**. ■

Lemma 7 is where the work is, and it is **prior art**:

- **Queyranne–Tardella Theorem 11** states it for the sublattice hull directly, and their **Theorem 3**

gives the projection representation under the sufficient condition "**I is finite**" — i.e. finitely many coordinates. Their proof is direct: fix x , choose $y^{\{ij\}} \in LQ$ agreeing with x on coordinates i, j , form the meets $u^i = \bigwedge_j y^{\{ij\}}$, then the join $z = \bigvee_i u^i$, and show $z = x$.

- **Baker–Pixley** gives the same conclusion abstractly: lattices have the majority term

$m(x,y,z) = (x \wedge y) \vee (y \wedge z) \vee (z \wedge x)$, and a variety with a majority term has every subalgebra of a finite product determined by its 2-fold projections (**k-decomposability**).

Either route suffices. **At $d = 2$ no lift is needed** — Lemma 5 is the whole proof.

Status: PRIOR ART, READ. EXHAUSTIVE cross-check: **36,252** cases over observed boxes, 0 failures; Lemma 7's conclusion checked directly on **4,128** generated sublattices, 0 exceptions, majority term valid on every one.

4.4 The correspondence, term by term

QUEYRANNE–TARDELLA	HERE
$\delta^Q_{ij}(h) = \bigvee \{x_i : x \in Q, x_j \leq h\}$	φ_{ij} (D4)
Proposition 1, $\pi_J LQ = L \pi_J Q$	Lemma 6
Theorem 9(ii), factors are chains	Lemma 5
Theorem 11, $LQ = \bigcap \text{Cyl}_{ij} E_{ij} Q$	Clause B
Example 10, failure off the chains	§8, N2
proper boundary epigraph: $k \geq \delta$ where attained, $k > \delta$ where not	§8, N1 — the hypothesis we needed and they did not

The last row is the instructive one, and an earlier draft overstated it. Their Remark 6 says that when each T_i is a chain **and Q is finite**, P^Q_{ij} is everything and the proper epigraph collapses to the plain one — so in *our* setting their strictness device is inactive too, and both formulations rely on the same finiteness-and-chains footing. What is true is narrower: **their formulation states the attainment condition explicitly and ours smuggled it into D1**, which is why our necessity analysis took two attempts and theirs needed none.

§5 • Clause C — **information is the join-closure**

C. $\text{op_information}(X) = J(X)$, and therefore $\text{information} \subseteq \text{algebra}$.

`op_information` extracts the join-irreducibles of X and re-closes them under $\vee$.

Lemma 8. Every $x \in X$ is a join of join-irreducibles of X . *Proof.* Induction on $n(x) = |\{y \in X : y < x\}|$. If x is join-irreducible it is its own seed. Else the supremum of $\{y \in X : y < x\}$ equals x , so x is the join of those elements, each with $n(y) < n(x)$, each therefore a join of seeds by induction. ■

Hence the seed regrows X , so $\text{op_information}(X) \supseteq X$; being join-closed it contains $J(X)$; and being generated by a subset of X it is contained in $J(X)$. So $\text{op_information}(X) = J(X)$. A sublattice is join-closed, so $J(X) \subseteq \langle X \rangle$. ■

Status: PROVED. SAMPLED: 200/200 positions, and 0 failures over the 600 worlds of the Clause H sweep (§6d).

`information` $\subseteq$ `algebra` is therefore a **theorem**, not the tendency it was recorded as.

§6 · Clause D — 2-determinacy, and where the content actually is

D. $L(X)$ is 2-determined **iff** it is pair-definable. Four of the five are; `information` is not.

($\Leftarrow$) Let $L(X) = \{x \in \text{Box} : (x_i, x_j) \in C_{ij} \ \forall i < j\}$. Then $\pi_{ij}(L(X)) \subseteq C_{ij}$, so $\{x : (x_i, x_j) \in \pi_{ij} L(X) \ \forall i < j\} \subseteq L(X)$; the reverse inclusion is free. ■ **Two lines.**

($\Rightarrow$) If $L(X)$ is 2-determined then $C_{ij} := \pi_{ij}(L(X))$ witnesses pair-definability. ■

The **iff** is true and nearly vacuous in the $\Rightarrow$ direction, and saying so matters: `order` (D5), `geometry` (D7) and `statistics` (D8) are each *defined* in pair form, so their 2-determinacy is free. **Measuring it measured nothing** — a correction to an earlier reading in this work that treated 500/500 across four operators as evidence.

The content is in exactly two places.

1. `algebra` is 2-determined and is not pair-defined. It is defined by closure under operations. Its 2-determinacy is precisely Lemma 7 — Theorem 11 / Baker–Pixley. **This is the whole substance.**
2. `information` is not 2-determined, minimal witness at $d = 3$:

$X = \{(0,0,0), (0,1,1), (1,0,1)\}$. $J(X)$ adds $(1,1,1)$ and stops at 4 cells. Rebuilt from its own pairwise projections, $J(X)$ **invents** $(0,0,1)$. On the same X , all four other operators are 2-determined.

One three-element set separates `information` from the rest of the hierarchy. The reason is structural: a **join-semilattice has no majority term**, so nothing in the Baker–Pixley family reaches it. Consistent with this, `information` still fails at $k = 3$ (270/300).

Status: PROVED (both directions, plus the witness). **SAMPLED:** 150/150 worlds for the four, failure confirmed for `information`.

§6b · Clause F — the order language is a precondition, and one language escapes it

F. Four of the five operators are **not invariant** under a relabelling of values that preserves their identity and destroys their order. `statistics` is invariant. **The order language is a precondition for** `order`, `algebra`, `information` **and** `geometry`, **and is not one for** `statistics`.

This clause exists because of a reading made in review: "*language is order. If order is necessary for algebra, then so is the language.*" That is right, and it reassigns the emphasis of §8 — see N1/N2 below.

The test. Take `X` in its box. Apply a bijection `σi` to each coordinate's observed values — pure relabelling, so every value keeps its identity and every membership fact is preserved, while the order `≤` is scrambled. Compute the operator on `σX`, map back by `σ-1`, compare with the operator on `X`.

OPERATOR	INVARIANT	VERDICT
<code>order</code>	58 / 400	requires the order
<code>algebra</code>	58 / 400	requires the order
<code>information</code>	59 / 400	requires the order
<code>geometry</code>	102 / 400	requires the order
<code>statistics</code>	400 / 400	ORDER-FREE

(The survivors are the draws where the shuffle happened to preserve the order, or the structure was degenerate. `geometry` survives more often for a reason F.2 makes exact, and it is not the one an earlier draft gave: not that a 2-D hull survives order-reversal, but that a hull survives every **affine** relabelling, in every dimension.)

`statistics`' **invariance is a theorem, not luck**. D8 reads only the membership `πij(x) ∈ πij(X)`, and every bijective relabelling preserves membership. Two lines. **PROVED**.

And the other four are order-dependent by explicit witness, not merely by inspection. Minimal counterexamples, found by exhaustive search over `d ≤ 3` and alphabets `≤ 4`:

OPERATOR	<code>X</code>	Σ	$ L(X) \rightarrow \Sigma^{-1}L(\Sigma X) $
<code>order</code>	<code>{{(0,0),(1,1)}}</code>	swap coordinate 2	<code>2 → 4</code>
<code>algebra</code>	<code>{{(0,0),(1,1)}}</code>	swap coordinate 2	<code>2 → 4</code>
<code>information</code>	<code>{{(0,0),(1,1)}}</code>	swap coordinate 2	<code>2 → 3</code>
<code>geometry</code>	<code>{{(0,0),(0,1),(1,2)}}</code>	<code>1↔2</code> in coordinate 2	<code>3 → 4</code>

Two cells suffice for three of the four. `{{(0,0),(1,1)}}` is a chain; relabel one coordinate and it becomes the antichain `{{(0,1),(1,0)}}`, whose sublattice closure is the whole box. **Order-dependence in two cells.** **PROVED** by witness, not by construction-inspection.

And they break on exactly the same instances — 400 / 400. An earlier draft called this an *independent* corroboration of Clause B. **It is not independent: it is entailed by it.** "`order` breaks on `X`" is `σ-1R(σX) ≠ R(X)` and "`algebra` breaks on `X`" is `σ-1{σX} ≠ {X}`; Clause B says `R(Y) = {Y}` for every `Y`, so with both `X` and `σX` in range the two conditions are literally the same condition. **The 400/400 is a consistency check, not evidence for the theorem.**

What the clause does and does not support. The claim under review was that *every rung above the first requires the language below it*. For algebra, information and geometry that is measured and true. For statistics it is **false** — and statistics is the rung the ladder placed at the top. Necessity of language is a condition of order for four of the five; the fifth needs only identity.

Status: PROVED — statistics from D8 in two lines, and the other four by the explicit witnesses above. The 400-draw sweep is **SAMPLED** corroboration, not the proof; an earlier draft filed the four as EXHAUSTIVE, which they were not.

F.2 · The refinement — "relabelling" is three operations, and the four are four groups

The clause above is true and it is coarse. **A relabelling is not one operation**, and the four operators that "require the order" do not require the same thing. Three relabellings live inside the word:

- **the full reversal** — every coordinate reversed at once, which is the **lattice dual**;
- **a mixed reversal** — some coordinates reversed and not others;
- **a permutation** — an arbitrary bijection of each coordinate's values, which is the test above.

Measured on the *same* four hundred draws as the table above, so the last column is that table's column unchanged and the two new ones are directly comparable to it:

OPERATOR	FULL REVERSAL	MIXED REVERSALS	PERMUTATION
order	400 / 400	78 / 1,564	58 / 400
algebra	400 / 400	78 / 1,564	58 / 400
geometry	397 / 400	1,555 / 1,564	102 / 400
information	40 / 400	111 / 1,564	59 / 400
statistics	400 / 400	1,564 / 1,564	400 / 400

(A draw of arity d contributes one full reversal and $2^d - 2$ mixed ones; the identity is excluded as trivial. The mixed totals are therefore 1,564 rather than 400.)

Every entry is accounted for, and the account is the refinement: **each operator is invariant under its own group of relabellings, the four groups are different, and where all four are determined they are strictly nested — in an order that is not the containment hierarchy.**

F.2. Let $G(L)$ be the set of coordinate relabellings under which L is invariant at **every** index, and let $R = \{id, rev\}^d$ be the reversal group — each coordinate independently reversed or not — with rev its all-reversed element, the lattice dual. Then

$G(\text{statistics})$ is everything; $G(\text{geometry}) \supseteq R$ on evenly-spaced boxes and excludes some permutation; $G(\text{order}) = G(\text{algebra}) = \{id, rev\}$ exactly; and $G(\text{information}) \cap R = \{id\}$.

So on R , where all four are determined, the four are strictly nested: $R \supsetneq \{id, rev\} \supsetneq \{id\}$, with statistics above R and geometry at R .

statistics — **the full symmetric group**. Already proved above: D8 reads only membership, and every bijection preserves membership. This is the whole of Clause F as originally stated.

order and algebra — the diagonal, and this is a theorem. Reversing every coordinate of a product of chains is an order-**antiautomorphism**: it inverts $\leq$, so it exchanges meet with join. The sublattice hull is closed under *both*, hence carried to itself, so $\sigma^{-1}(\sigma X) = \langle X \rangle$ for the full reversal at every index — which is the 400 / 400 above, and it is exact rather than a rate. **PROVED.** A *mixed* reversal is neither an automorphism nor an antiautomorphism of the product, and the two operators break under it: the witness is already in the table above, since swapping the two values of one coordinate of $\{(0,0), (1,1)\}$ is a mixed reversal. So $G(\text{order})$ contains the diagonal and no mixed pattern, and the 78 / 1,564 is coincidence. Nothing further can be added either: **order** and **algebra** read the index only through $\leq$, and the only bijections of a finite chain that preserve or invert its order are the identity and the reversal. Hence $G(\text{order}) = G(\text{algebra}) = \{\text{id}, \text{rev}\}$, **exactly**, and this is the one of the four groups pinned from both sides.

geometry — the reversal group, once the encoding is affine. A convex hull commutes with any affine bijection of the coordinates, and each coordinate may be reversed independently, so the whole group $\{\text{id}, \text{rev}\}^d$ is admitted — not merely the diagonal, and not only at $d = 2$. The qualification is real and is the reason for the three misses: relabelling acts here by **rank**, and a rank reversal is affine exactly when the coordinate's observed values are equally spaced. Split by that condition, across all 1,964 reversals of the sweep:

	REVERSALS	GEOMETRY MISSES
equally-spaced boxes	1,382	0
otherwise	582	12

Every miss is a non-affine encoding and none is a failure of the hull. **geometry** is not invariant under a general permutation — the third witness in the table above — so its group is the reversal group and no more.

information — no reversal at all, and the witness is the sharpest of the four. **information** is the join-closure (Clause C), and the dual exchanges join with meet, so the dual sends it to the *meet*-closure — a different operator. The minimal counterexample is two cells:

$X = \{(0,1), (1,0)\}$, the two-cell antichain, under the full reversal. $\text{information}(X)$ is the join-closure $\{(0,1), (1,0), (1,1)\}$; pulled back through the dual it is the meet-closure $\{(0,1), (1,0), (0,0)\}$. The other four operators are all invariant here.

Two features of that witness matter. First, **X is self-dual as a set** — the reversal maps it to itself — so the failure is not that the index moved. Nothing moved; the *operator* is not dual-equivariant. Second, **the two closures have the same size, three and three**. A test that compared cardinalities would have reported invariance. This is the same discipline as §7's refusal to read a size comparison as a containment, and here it is load-bearing: the 40 / 400 is unreachable by counting.

Together with the mixed-reversal witness already in the table, that rules out **every** non-identity element of R , which is what the chain needs: $G(\text{information}) \cap R = \{\text{id}\}$. Whether some exotic permutation of a larger alphabet survives at every index is **not settled here**, and the boxed statement is confined to R for that reason.

What the refinement changes. Clause F says $G(\text{statistics})$ is strictly the largest of the five. That is true, and it is the weakest of four separate facts. It does **not** say the other four behave alike, and they do not: **geometry** survives a reversal **order** cannot, **order** survives a reversal **information** cannot, and **information** survives none. The clause stands; its four-and-one partition is a partition of $G(\text{statistics})$ against the rest and not a description of the rest.

Status: PROVED — `statistics` from D8; `order` and `algebra` from the antiautomorphism argument; `geometry` from affine-invariance of the hull, with the rank-versus-affine condition stated rather than absorbed; `information` refuted at the dual by the two-cell self-dual witness. Each non-containment in the chain carries an explicit minimal witness. **The three sweep columns are SAMPLED corroboration on 400 draws (1,564 mixed reversals), not the proof.**

§6c · Clause G — `statistics` is the hypothetical language, and that settles N1

G. `statistics` is the only operator whose data is *cells* rather than *relations between cells*. Its admission condition is a membership test, so it is order-free; it lies inside `geometry` always; and its hypothetical extension preserves the geometry that generated it.

This clause exists because of a reading offered in review: *"Statistics remains without order because it is the only language that speaks in hypotheticals. Each other language requires order. Statistics only needs a singularity."* Each part is measurable and each part holds.

G.1 — "it only needs a singularity." $\text{St}(X)$'s constraint set is $\pi_{ij}(X) = \bigcup_{y \in X} \{\pi_{ij}(y)\}$ — a **union of single-cell contributions**. No relation between two cells is ever consulted.

UNION OF SINGLE-CELL FACTS	
<code>statistics</code>	500 / 500
<code>algebra</code> (<i>control</i>)	98 / 500

The control matters: the property is not generic. `algebra` needs $\wedge$ and $\vee$, which are relations *between* cells; `statistics` needs only the cells themselves. **PROVED** (immediate from D8), and **SAMPLED** at 500 worlds.

G.2 — the hypothetical character and the order-freedom are the same fact. A membership test asks *"has this pair been seen?"* — an `if`, evaluated against observation. Membership is preserved by every bijection, so Clause F's 400/400 order-freedom is not a second property; **it is G.1 restated**. The other four read $\leq$ directly ($\emptyset$, $\wedge$, $\vee$, `conv`) and cannot be evaluated without it.

G.3 — `statistics` $\subseteq$ `geometry`, always. 500 / 500, and it is a one-line proof: $\pi_{ij}(X) \subseteq \text{conv}(\pi_{ij} X)$, so any cell whose pair-projections are *observed* has them in the *hull*. **PROVED**. This is the second lawful containment after Clause C.

G.4 — the hypotheticals never leave the geometry that generated them.

`geometry(statistics(X)) = geometry(X) — 500 / 500.`

$X \subseteq \text{St}(X) \subseteq G(X)$ by G.3, and `conv` of any set between S and `conv(S)` is `conv(S)`, so the hulls are unchanged. **PROVED**. This is the measurable part of the review's third sentence — *"logic verifies the if and then against the original geometry being statisticized."* **The verification step holds.**

G.5 — and this settles N1. N1 and N2 are stated in different *registers*, and the register decides which one can be necessary:

HYPOTHESIS	INVARIANT UNDER AN ORDER-DESTROYING RELABELLING	REGISTER
N1 every ambient value observed	500 / 500	membership
N2 the order relation itself	84 / 500	order

And the tempting inference from this is invalid, so it is not made. An earlier draft argued "a condition stated in the membership register cannot be necessary for an order-theoretic theorem" — that does not follow. Invariance mismatch blocks *equivalence*; it does not block one-way implication, and an order-invariant condition can perfectly well imply an order-dependent one. Indeed N1 does exactly that: it implies attainment (Lemma 1) without stating it.

What the register table does establish is weaker and still useful: N1 and N2 are conditions of different kinds, so the fact that one is necessary and the other is not is **not a coincidence of counterexamples but a difference in what they can express**. The necessity itself is settled by Lemma N1* and its corollary in §8, not by this table.

G.6 — what is NOT testable here, stated so it is not smuggled in. The review's chain runs *statistics gives the* if → *analysis gives the* then → *logic verifies both against the original geometry*. The first and third steps are measured above and hold. **The middle step cannot be measured**: no operator is given for *analysis* anywhere in this paper, and which operators belong to the family at all is a question this paper does not settle. **No operator for *analysis* was invented for this clause**, and none should be invented until that question is settled on its own terms.

§6d · Clause H — the lawful skeleton: which part of the ranking is a law

Clause E refutes the *total* ranking. It does not follow that no ordering is lawful, and review pressure made that worth measuring. Over 600 worlds, exactly **seven** containments hold without exception:

HOLDS ALWAYS	
order ⊆ algebra and algebra ⊆ order	= Clause B, recovered from outside its own derivation
information ⊆ algebra, information ⊆ order	Clause C
statistics ⊆ algebra, statistics ⊆ order	proved below — the fourth relation, which an earlier draft left with a blank cell
statistics ⊆ geometry	G.3

Everything else varies — *geometry ⊆ order* 585/600, *statistics ⊆ information* 422/600, *information ⊆ geometry* 301/600, and so on down to 139/600.

The lawful skeleton, with 0 := order = algebra:

0 = order = algebra	geometry
/	\
information	statistics —

- 0 and geometry are **incomparable** — neither contains the other in general.
- information and statistics are **incomparable** with each other.
- statistics is the only language below **two** maxima, and the only one below geometry.
- information and statistics are the **minimal** elements; 0 and geometry the **maximal**.

The fourth lawful relation, `statistics ⊆ order = algebra`, has a three-line proof and an earlier draft cited three clauses for four relations, leaving this one unjustified:

Let $x \in \text{St}(X)$. For each pair $i \neq j$ there is, by D8, some $y \in X$ with $(y_i, y_j) = (x_i, x_j)$. That y satisfies $y_j \leq x_j$ and $y_i \geq x_i$ — with equality — so it is exactly the witness Clause B's form of $R(X)$ requires. Hence $x \in R(X) = \langle X \rangle$. ■

Each incomparability is PROVED by an explicit minimal witness, not merely measured:

NON-CONTAINMENT	X	WITNESS CELL
<code>geometry ⊄ order</code>	<code>{(0,0),(0,1),(1,2),(2,2)}</code>	<code>(1,1)</code>
<code>order ⊄ geometry</code>	<code>{(0,1),(1,0)}</code>	<code>(0,0), (1,1)</code>
<code>information ⊄ statistics</code>	<code>{(0,1),(1,0)}</code>	<code>(1,1)</code>
<code>statistics ⊄ information</code>	<code>{(0,0,0),(0,1,1),(1,0,1)}</code>	<code>(0,0,1)</code>
<code>information ⊄ geometry</code>	<code>{(0,1),(1,0)}</code>	<code>(1,1)</code>
<code>geometry ⊄ information</code>	<code>{(0,0),(0,2),(1,1)}</code>	<code>(0,1)</code>

The two-cell antichain `{(0,1),(1,0)}` separates three of the six. And the `statistics ⊄ information` witness is the same three-cell set that breaks 2-determinacy in Clause D — one object doing both jobs, which is why `information` is the odd language in two independent senses.

This is the precise sense in which the ladder was right and wrong. It was wrong as a total order (Clause E). It was right that there is a hierarchy: a genuine partial order with four lawful relations, in which `statistics` sits at a minimum — the most restrictive position, which in the ladder's own idiom is the top rung. Status: SAMPLED (600 worlds); the four relations are each individually PROVED (Clause B, C, G.3).

§7 • Clause E — the ranking is not part of the law

E. The total order in which the five languages nest is a property of the index, not a law. (Clause H gives the part that is lawful — read them together.)

The ranking this clause refutes was read off a single index: 17 energy conditions over five coordinates in a 288-cell box. It is a real index and the ranking really does hold on it — which is precisely why the clause is needed. Measured over 400 random worlds:

OBSERVATION	VALUE
distinct size-orderings of the five	14
frequency of that index's ordering among the 400	42 / 400 (fourth most common)
statistics is the minimum (below all four others, including information)	106 / 400
statistics is the maximum	128 / 400
geometry vs information	info < geom 183 · incomparable 144 · equal 57 · geom < info 16
order is the maximum	398 / 400 (both exceptions geometry, both incomparability)

These two figures measure different things and an earlier draft let them sit fifty lines apart unreconciled. "statistics is the minimum" here means below all four other languages — which fails whenever information is incomparable with it. Clause H's statistics $\subseteq$ order, $\subseteq$ algebra, $\subseteq$ geometry hold always; it is only statistics vs information that varies. 106/400 is the four-way count; the three lawful containments are 600/600.

And two of the indexes studied here already disagree — but on size, not on containment, and an earlier draft conflated the two. (The energy-condition figures below moved from 29 and 156 when an arity coordinate was added to that index: one of its cells had been seated under a name whose condition is bilinear, which the index's declared quadratic form cannot express. Both the incomparability and the direction of the size difference are unchanged by the correction.)

INDEX	GEOMETRY	INFORMATION	SET RELATION
the energy-condition index above	30	208	INCOMPARABLE
a synthetic periodic index in three coordinates	173	114	INCOMPARABLE

△ THE "SWAP" IS A CARDINALITY ARTEFACT. geometry and information are set-theoretically incomparable on both seated indexes — neither contains the other — so nothing nests one way on one index and the other way on the other. What differs is only which set is larger. The earlier draft presented the size order as a nesting order and called the reversal a swap; that reading is withdrawn. The clause survives on the sampled evidence above, and on the fact that a size ranking is not an order relation at all — which is itself the point.

Status: REFUTATION, SAMPLED (400 worlds). The ranking was a property of that one index.

§8 · Necessity — each hypothesis dropped until it breaks

The two hypotheses are not peers, and review corrected this. N2 is the order language being present and total. N1 is a bookkeeping condition about attainment. The reading offered was: "language is order; if order is necessary for algebra, so is the language." Clause F measures exactly that — scramble the order and algebra changes in 342 of 400 draws. So the necessary hypothesis of Clause B is the language (N2), and N1 is not necessary at all. They are listed in that order below and the emphasis has been moved accordingly.

N1 · $A_i = \pi_i(X)$ (observed alphabet) — SUFFICIENT, NOT NECESSARY, AND THE SHARP CONDITION IS NOW PROVED.

Lemma N1*. For any declared box $B \supseteq \text{Box}(X) : R_B(X) \cap \text{Box}(X) = \langle X \rangle$. *Proof.* φ_{ij} is computed from X alone and does not mention B . So for $x \in \text{Box}(X)$ the staircase conditions over B are literally the same conditions as over $\text{Box}(X)$, whence $R_B(X) \cap \text{Box}(X) = R_{\{\text{Box}(X)\}}(X)$, which is $\langle X \rangle$ by Clause B. ■

Corollary (necessary and sufficient). $R_B(X) = \langle X \rangle$ iff $R_B(X) \subseteq \text{Box}(X)$. The failure set is exactly $R_B(X) \setminus \text{Box}(X)$ — the cells outside the observed box that survive the staircase. **PROVED**, and EXHAUSTIVE at **0 failures in 702,628 cases**.

An earlier draft recorded N1 as necessary. That was wrong, and the correction has two stages, both kept because the second supersedes the first rather than erasing it.

First correction. Over a declared box larger than the observed one the identity often fails but **not always**, so N1 is sufficient and not necessary. A weaker sufficient condition is **order-convexity** — no value of A_i strictly between $\min \pi_i(X)$ and $\max \pi_i(X)$ may go unobserved, while values above the max or below the min are harmless. Measured over declared boxes: **convex** → **1,853 hold, 0 fail; gapped** → **632 hold, 1,643 fail**. The 632 show convexity is *also* not necessary. Minimal counterexample: $d = 2$, box of 3 cells, $X = \{(0,0), (2,0)\}$ with value 1 unobserved — staircase 3 cells, sublattice 2.

Second correction, which supersedes it. **Convexity is not the boundary; the corollary is.** A gapped world holds precisely when no outside cell survives the staircase, and convexity is sufficient exactly because it forces that. The 632 are not an anomaly to be explained away — they are the corollary being weaker than convexity, as it should be.

And Queyranne–Tardella need no such hypothesis, though not for the reason first given here: their Remark 6 has the proper epigraph collapse to the plain one when the factors are chains and Q is finite, so the strictness device is inactive in our setting too. What their formulation does is **state the attainment condition explicitly** where ours buried it in D1.

N2 · each A_i a chain — NECESSARY. THIS IS THE LANGUAGE CONDITION, and the failure is one-directional. Off the chains the staircase **never over-generates**; it **under-generates**. $R(X) \subseteq \langle X \rangle$ survives arbitrary finite lattice factors — it is Lemma 4, which needs φ isotone and hence a total order, that dies. Measured over non-chain factors:

The ambient must be stated, because off the chains it is load-bearing and the counts move with it. A subset of a non-chain factor need not be a sublattice of it, so D2's justification fails and there are two defensible ambients: the **observed** product $\prod \pi_i(X)$, and the **closed** product in which each factor's observed values are closed under that factor's own meet and join.

FACTORS	OBSERVED AMBIENT — EQ / UNDER / OVER	CLOSED AMBIENT — EQ / UNDER / OVER
chain × chain	120 / 0 / 0	120 / 0 / 0
M3 × chain	290 / 270 / 0	356 / 204 / 0
M3 × M3	700 / 1900 / 0	1066 / 1534 / 0
N5 × chain	371 / 189 / 0	415 / 145 / 0
N5 × N5	1144 / 1456 / 0	1428 / 1172 / 0

Every equality and under-generation count changes with the ambient. The over-generation column does not: it is 0 under both. So the conclusion this section draws — that the staircase *under-generates* off the chains and never over-generates — is ambient-independent, while the individual counts are not. An earlier draft printed one ambient's numbers without naming the convention.

Queyranne–Tardella's **Example 10** is the same phenomenon on $\{0, a, b, c, 1\}$ with $a \wedge b = 0$, $a \vee b = c < 1$.

N2a · the proper-epigraph formulation does NOT rescue the non-chain case — tested, and it is a proof rather than a measurement. Queyranne–Tardella's *proper* boundary epigraph distinguishes attained from unattained boundary values, admitting $k \geq \delta$ in the first case and requiring $k > \delta$ in the second. The obvious hope was that this strictness is what chains were standing in for. **It is not.** Measured over non-chain factors against the naive staircase:

FACTORS	CASES	NAIVE: EQ / UNDER / OVER	PROPER: EQ / UNDER / OVER
chain × chain	120	120 / 0 / 0	120 / 0 / 0
M3 × chain	560	290 / 270 / 0	290 / 270 / 0
M3 × M3	2600	700 / 1900 / 0	700 / 1900 / 0
N5 × chain	560	371 / 189 / 0	371 / 189 / 0
N5 × N5	2600	1144 / 1456 / 0	1144 / 1456 / 0

Identical to the case. And the reason is a one-line proof, not a coincidence: strictness can only *remove* admitted cells, so $\text{proper}(X) \subseteq \text{naive}(X)$ always. Off the chains the failure is **under-generation** — $\text{naive}(X) \not\subseteq \langle X \rangle$ — so no operator contained in the naive one can reach $\langle X \rangle$. **Attainment-strictness cannot repair under-generation.** The chain hypothesis is doing independent work: it is what makes Lemma 4 ($\langle X \rangle \subseteq R(X)$) true, and Lemma 4 is the inclusion that dies.

N3 · $d \geq 2$ — a convention. `op_order` returns SILENT at $d = 1$; both sides would be X .

N4 · finiteness — it is the number of factors that Lemma 7 needs, not finiteness of the factors. Queyranne–Tardella's Example 2 (finite subsets of an infinite index set) shows the representation failing when I is infinite without a subcompleteness condition.

§8b · Appendix — analysis, the residual, and why logic is needed

WHICH OPERATORS BELONG TO THE FAMILY IS NOT SETTLED HERE. The five of Clauses A–H are unchanged and nothing below adds a sixth. This is a candidate operator, specified in review, measured, and filed as **evidence**.

△ THE FIRST VERSION OF THIS APPENDIX IS WITHDRAWN. It formalized "best possible closure route" as *minimum number of $\wedge / \vee$ steps* and concluded that `logic` had almost nothing to do — only 0.3 % of geometric positions unreachable. **That measured REACHABILITY, which is not closure,** and the conclusion was wrong by a factor of two hundred. The corrected reading and measurement follow.

The specification, corrected in review. " $E = |X|$ means not all statistical positions can close at $E = 0$." So $E = |X|$ is not a value to hit; it is the assertion that **the residual is not always zero**. `analysis` reports, for each statistical

position, the **best achievable residual** — and because that residual is often nonzero, something must adjudicate it. **That is logic**.

Formalization. For each statistical position $x \in \text{St}(X) \setminus X$, form $Y = X \cup \{x\}$ and compute $E_L(Y) = |L(Y)| - |Y|$ for each language L . The *best closure route* for x is the L minimising $E_L(Y)$; **analysis** reports that minimum. x **closes** iff the minimum is 0.

Measured over 155 statistical positions in 79 worlds:

reachable — a finite route exists	100 % (theorem: statistics $\subseteq$ algebra , Clause H)
closes at E = 0 by its best route	45 / 155 — 29 %
CANNOT close at E = 0 by any route	110 / 155 — 71 %
worlds containing at least one such position	34 / 79

Distribution of the best achievable residual: {0:45, 1:36, 2:26, 3:3, 4:11, 5:26, 7:8}.

REACHABLE BUT NOT CLOSABLE. Every statistical hypothetical lies in the sublattice hull, so a route always exists — and yet **71 % of them cannot be closed to E = 0 by any language**. The two properties are independent, and conflating them is what produced the withdrawn version above.

And the best route is always statistics. Per language, the fraction of positions reaching $E = 0$:

LANGUAGE	AT E = 0
order	2.6 %
algebra	2.6 %
information	7.1 %
geometry	8.4 %
statistics	29.0 %

The best-over-all-languages figure is **also 29.0 %** — identical to **statistics** alone. **Whenever any language can close a position, statistics can.** But it is **not** true that no other language ever beats it, and §8c's own routing table shows why: **information alone is the best route on 13 positions**, where it reaches a strictly smaller residual than **statistics** does. The correct statement is the one about reaching zero, not about the residual in general.

statistics is optimal for E = 0 and not optimal in general. An earlier draft asserted the stronger claim and the paper refuted it sixty lines later without noticing.

What this settles about analysis. It returns a **residual magnitude**, not a cell decision, and the magnitude is nonzero for 71 % of its domain. So it has a mechanism but does not decide a cell, which is the criterion the five satisfy and it does not — now with the mechanism named and the magnitude measured. **analysis is not a sixth operator-bearing language**, and the 71 % is exactly why the chain needs a third step: **statistics** proposes the **if**, **analysis** returns the residual, **logic** **adjudicates the residual that analysis declines to judge**.

Status: SAMPLED, CANDIDATE. The draw is seed 53 over 300 worlds — **d** in 2–4, alphabets 2–4, $|X|$ in 2–7 — which yielded the 155 statistical positions in 79 worlds reported here.

§8c · Appendix — `logic` is an interpreter, not a language, and its refusals are seed-dependent

WHICH OPERATORS BELONG TO THE FAMILY IS NOT SETTLED HERE. Candidate, measured, filed as evidence. The five of Clauses A–H are unchanged.

⚠ **THIS APPENDIX'S FIRST HEADLINE IS WITHDRAWN.** It reported "*8 of 155 positions refuted*" and concluded that **the residual does not determine the verdict**, calling that the argument for `logic` being a distinct step. **All 8 refutations were tie-break artifacts.** When several languages tie at the minimum residual, "the best closure route" does not name a unique closure, and the tie was broken by enumeration order, which is arbitrary — two passes over the same worlds gave 147 and 152. The claim rested entirely on those 8 and does not survive. Corrected below.

The specification, refined in review. "*Logic is not exactly a language. It is an interpreter. It interprets the statistical analysis into a binary verdict. So logic is multilingual.*" That dissolves rather than settles the earlier tension: the previous version applied **language criteria** (Clause A's closure axioms) to something that is **not a language**, and unsurprisingly found a contradiction.

The obvious reading is empty. Asking whether the *position* lies in `G(X)` is 155 / 155 — that is just Clause G.3. Logic must read the closure `analysis` produced.

The verdict, bracketed — because the tie-break is not part of the specification:

RULE	ADMITTED
unanimous — every minimizing language admits	147 / 155 (94.8 %)
existential — some minimizing language admits	155 / 155 (100 %)
the gap — verdict depends on which tied language is read	8 (5.2 %)

The gap sits at residuals `{2: 2, 4: 3, 5: 3}`. Languages tie at the minimum in 35 of 155 positions (2-way 16, 3-way 13, 5-way 6); only 120 have a unique best route.

AT SEED 53, `logic` never refutes a position outright — every one of the 155 is admitted by at least one minimizing language.

⚠ **AND THAT DOES NOT GENERALISE.** At seed 77 the same sweep gives **21 refusals in 177 positions**. The universal reading — "there is no language-independent rejection" — was asserted in an earlier draft on the strength of one seed and is **WITHDRAWN**. What survives is the weaker and still interesting fact that outright refutation is *rare* and *seed-dependent*, which is itself evidence that the verdict is not a property of the position alone.

And that is what makes it an interpreter. Its output is **not a language-independent fact about the position** — it is a verdict *relative to a language*, and where languages tie it must arbitrate between them. A closure operator returns a set; a language returns a binary about a cell; **an interpreter returns a binary about a claim made in some language, and**

must say which language it read. The 8 gap positions are the multilingualism made visible: same position, same residual, different language, different answer.

Which language does analysis route through?

BEST ROUTE	POSITIONS
statistics alone	107
information alone	13
geometry = information = statistics	13
geometry = statistics	11
all five tied	6
information = statistics	5

statistics is in the winning set for **142 of 155**, consistent with Clause G — but it is not alone in **35**, and that is where an interpreter is required.

Forced monolingual, the verdict swings wildly: order 19/155 · algebra 19/155 · information 67/155 · geometry 155/155 · statistics 155/155. **A single language cannot stand in for the interpreter** — reading only order would reject 88 % of what reading only geometry accepts.

What this settles about Clause A. Nothing. logic is not a language, so the closure axioms were never the right test, and the earlier "passes the binary criterion, fails Clause A" tension was an artifact of the wrong frame. **Clause A stands unchallenged;** logic is simply outside its scope.

Status: SAMPLED, CANDIDATE. Same draw as §8b: seed 53 over 300 worlds. The figures below are properties of that draw, not of the operators — the seed-77 counterexample above is the proof of it. The tie-break remains unspecified; any figure quoted without saying *unanimous* or *existential* is unstable across an 8-position band.

§8d · The generalized theorem

The law as stated is one point in a three-axis family. This section names the axes, says what survives each weakening, and marks what is open.

The abstract form. Let V be a variety with a $(k+1)$ -ary **near-unanimity term**, let $B = \prod_{i \in I} A_i$ be a product of algebras of V over a **finite** index set I , and let $Q \subseteq B$. Then the subalgebra generated by Q is **k-decomposable**:

$$\langle Q \rangle = \{ x \in B : \pi_S(x) \in \pi_S \langle Q \rangle \text{ for every } S \subseteq I \text{ with } |S| = k \}.$$

Our law is the instance $V = \text{lattices}$, $k = 2$, majority term $m(x,y,z) = (x \wedge y) \vee (y \wedge z) \vee (z \wedge x)$, each A_i a finite chain — with the additional computational content, *not* supplied by the abstract form, that the 2-fold projections are given in closed form by the staircase φ (Lemma 5). **The abstract theorem says the projections determine the hull; Lemma 5 says what the projections are.**

The three axes

Axis 1 — the variety, i.e. the arity of the near-unanimity term. A $(k+1)$ -ary NU term buys k -decomposability. Lattices have a 3-ary one, hence pairs. **Join-semilattices have none** — an NU term forces congruence distributivity, which semilattices lack — and this is exactly why `op_information`, which is a join-closure (Clause C), escapes the whole family.

PROVED, for every dimension. Earlier drafts of this work reported the k -universality as open, on the strength of a run showing 188/188 at $k = 4$ — **a size artifact**, which vanished once $|X|$ grew (219/250 fails). The general statement is now a theorem with an explicit construction.

Proposition (information is k -determined for no non-vacuous k). For $d \geq 3$ put $X_d = \{0\} \cup \{e_i + e_d : 1 \leq i \leq d-1\} \subseteq \{0,1\}^d$. Then:

(a) $e_d \notin J(X_d)$. Every element of X_d is 0 or has at least two ones, and a coordinatewise join of such elements is 0 or has at least two ones; e_d has exactly one. ■

(b) For every T with $|T| = k \leq d-1$, $\pi_T(e_d)$ is a k -projection of an element of $J(X_d)$. If $d \notin T$ then $\pi_T(e_d) = 0 = \pi_T(0)$. If $d \in T$ then, since $|T| \leq d-1$, some $i \notin T$ with $i \neq d$ exists, and $e_i + e_d$ agrees with e_d off coordinate i , so $\pi_T(e_i + e_d) = \pi_T(e_d)$. ■

(c) Hence $J(X_d)$ is not k -determined for any $k \leq d-1$, and $k \geq d$ is vacuous. ■

Verified in every dimension $3 \leq d \leq 8$, with $|J(X_d)| = 2^{d-1}$ throughout. `information` **escapes the near-unanimity family entirely — not for small k , but for all of it.**

Axis 2 — the factors. The abstract form needs only $A_i \in V$. **Lemma 5 needs chains**, and that is where the law is sharp: off the chains $\langle X \rangle \subseteq R(X)$ fails while $R(X) \subseteq \langle X \rangle$ survives, so the staircase **under**-generates (§8, N2), and no attainment-strictness repairs it (§8, N2a). The abstract layer survives arbitrary factors in V ; **the computational layer does not.**

Axis 3 — the index set. $|I|$ finite is what Theorem 3(i) of the source requires. Dropping it breaks the representation — Queyranne–Tardella's Example 2 (finite subsets of an infinite index set) is a sublattice whose 2-fold projections are everything. Subcompleteness conditions (their (ii)) recover it. **The factors need not be finite; the index set does.**

What is general and what is ours

LAYER	STATEMENT	STATUS
abstract	NU term $\implies k$ -decomposability	PRIOR ART — Baker & Pixley 1975
representation	the hull is fixed by 2-fold projections, chains	PRIOR ART — Queyranne–Tardella Thm 11
computational	the projections are φ ; the four-witness construction	Lemma 5 — ours as written, theirs in substance
the five operators	Clauses A, C, D, F, G, H	ours

The honest summary of the generalization. Clause B generalizes cleanly along Axis 1 and Axis 3 and **not at all along Axis 2**. Clauses C, F, G and H are statements about five specific operators and do not generalize beyond them — they are facts about *these five operators*, not about closure operators at large.

§9 · Provenance ledger

ITEM	STATUS	SOURCE
B , $R(X) = \langle X \rangle$	PRIOR ART, READ	Queyranne & Tardella, <i>Discrete Math.</i> 308 (9) (2008) 1508–1523 — Prop 1, Thm 9(ii), Thm 11
the lift, abstractly	PRIOR ART, CITED-UNREAD	Baker & Pixley, <i>Math. Z.</i> 143 (1975) 165–174, DOI 10.1007/BF01187059
sublattices of finite products	PRIOR ART, CITED via Q–T, RESOLVED	Topkis [16, Thm 1] = D.M. Topkis, <i>The structure of sublattices of the product of n lattices</i> , Pacific J. Math. 65 (1976) 525–532. Bracket number read off Queyranne–Tardella's own bibliography, now transcribed in our reconstruction of their paper, §References. Not read in the original ; the theorem's content is taken from Q–T's statement of it
products of chains	PRIOR ART, CITED via Q–T, RESOLVED	Veinott [18, Cor 11] = A.F. Veinott Jr., <i>Representation of general and polyhedral subsemilattices and sublattices of product spaces</i> , Linear Algebra Appl. 114/115 (1989) 681–704. Stated there for <i>certain</i> sublattices of an arbitrary product of chains — the restriction is in Q–T's text and must not be dropped. Not read in the original , as above
$\langle X \rangle$ as sublattice closure	PRIOR ART	Birkhoff, <i>Lattice Theory</i> — already <code>op_algebra</code> 's citation
the majority term	PRIOR ART, STANDARD	$m(x, y, z) = (x \wedge y) \vee (y \wedge z) \vee (z \wedge x)$
Lemmas 1–3, Clause A	OURS, PROVED	§2–§3
Lemma 5, the four-witness construction	OURS AS WRITTEN, THEIRS IN SUBSTANCE	§4.2. Queyranne–Tardella Theorem 9(ii) is the same statement; the four-witness proof is written here, the result is not new. §4.4 maps them
N1 sharp (Lemma N1*)	OURS, PROVED + MACHINE-CHECKED	the necessary-and-sufficient corollary in §8; order-convexity is demoted to <i>sufficient for that corollary</i>
N2 one-directional failure off chains	OURS, EXHAUSTIVE	§8
Clause C	OURS, PROVED	§5
Clause D	OURS, PROVED	§6 — three of four cases trivial, and said so
Clause E	OURS, REFUTATION	§7

Naming corrections carried by the ledger. The property in Lemma 7 is called **k-decomposability**. It is *not* "near-unanimity" — that names the *term*, and a $(k+1)$ -ary near-unanimity term is what is *equivalent* to k-decomposability. It is *not* "skew-free" — that is the Fraser–Horn property about congruences of a product, a different statement that sits in the same textbook section.

On the reconstruction. Queyranne–Tardella was not available to us as publisher's text, and was read instead from photographs of the printed article: Sections 1–3, the start of 4, and the Acknowledgements, References and the Theorem 9 attribution endnote. Prose is close to verbatim, but **the mathematical notation was restored by hand and is a reading**, and two displayed formulas could not be recovered with confidence. **Every quotation of a formula in this paper should be checked against the published article before it is relied on.**

A provenance row records what we have read, never what exists. Both rows above stood at **UNRESOLVED** until their bracket numbers were read off Queyranne–Tardella's own bibliography, and neither the works nor their identities changed when they were resolved — only our access to them did. Alongside that bibliography sits the authors' endnote dividing credit for Theorem 9: condition (i) $\implies$ (2) is Topkis's Theorem 1 with their Proposition 1, or Veinott's Corollary 11 when every factor is a chain, and **condition (ii) they claim as new**. Condition (ii) is the clause this paper imports, and Queyranne–Tardella are its authors.

§10 · Verification record

EXHAUSTIVE and SAMPLED are different words here. A figure like "300/300" is a *seeded pseudorandom sweep*, not a decision procedure over a finite family, and an earlier draft filed both under EXHAUSTIVE. They are separated below. Every SAMPLED figure states the size and shape of the draw it came from; every EXHAUSTIVE figure names the family it exhausted.

OBJECT	PROVED	EXHAUSTIVE	SAMPLED (SEEDED)	MACHINE-CHECKED (Z3)
Lemmas 1, 2, 3	✓	—	400 nested pairs	—
Clause A, five operators	✓	—	300 worlds	—
Lemma 4	✓	—	—	✓ 3×3, 4×4, 2 ³ , 3×3×3
Lemma 5 ($d = 2$)	✓	54,392 cells, construction built	—	✓ via Clause B at $d = 2$
Lemma 7 (the lift)	cited	4,128 sublattices	—	—
Clause B	cited + ✓	36,252 cases	—	✓ 2×2, 3×3, 4×4, 2 ³ , 3×3×3
Clause C (engine, Lemma 8)	✓	—	600 worlds	✓ 3×3, 2 ³ , 3×3×3
Clause C ($\subseteq$ algebra)	✓	—	200 positions	✓ 3×3, 2 ³ , 3×3×3
Clause D	✓	—	150 worlds + explicit witness	—
Clause E	refutation	—	400 worlds	—
Clause F	✓ + 4 witnesses	—	400 draws	—
Clause F.2 (the four groups)	✓ + 3 witnesses	—	400 draws, 1,964 reversals	—
Clause G	✓ (G.1, G.3, G.4)	—	500 worlds; control 98/500	—
Clause H	✓ + 6 witnesses + the §6d proof	—	600 worlds	—
N1 (sharp, Lemma N1*)	✓	702,628 cases	—	✓ both directions , 3×3, 4×4, 3×3×3
N2 / N2a (non-chain)	✓ (direction)	6,440 instances × 2 ambients	—	—
§8d, information at every k	✓ (construction)	$d = 3..8$	—	—

The five exhausted families, named. §0 defines EXHAUSTIVE as a decision procedure that visited every case in a *stated finite family*, so the families are stated here rather than left as bare counts:

FIGURE	THE FAMILY IT EXHAUSTED
54,392 cells (Lemma 5)	every cell of $R(X)$, over every X with $ X \leq 5$ in a 3×3 and a 4×4 box
36,252 cases (Clause B)	3×3 all subsets; 4×4 and 3×3×3 with $ X \leq 4$; 2 ⁴ with $ X \leq 5$; 2 ⁵ with $ X \leq 3$
4,128 sublattices (Lemma 7)	every $\langle X \rangle$ for $ X \leq 3$ in a 3×3, a 3×3×3 and a 2 ⁴ box
702,628 cases (Lemma N1*)	$d \in \{2, 3\}$, every coordinate on 3 or 4 values, every X with $ X \in \{2, 3, 4\}$
6,440 instances (N2/N2a)	the five factor pairs tabulated in §8, each enumerated in full, under both ambients

21 of 21 machine-check obligations discharged (Z3 5.1.0), subject to the two conditions below, which must hold before any of them counts.

What machine-checking bought that enumeration could not

Obligations are quantified formulas whose variables range over **every** subset X of the box — and, where $\langle X \rangle$ appears, over **every** closed superset S . Z3 returns `unsat` on the negation.

At $d = 3$ over a $3 \times 3 \times 3$ box that is $2^{27} = 134,217,728$ subsets. The enumeration frontier there is $|X| \leq 4$.
Machine-checking covers cases enumeration cannot enumerate.

Two encodings make it decidable. φ and $\max$ are eliminated — $x \in R(X)$ iff for all $i \neq j$ there is $y \in X$ with $y_j \leq x_j$ and $y_i \geq x_i$, since a maximum is attained exactly when a witness exists. And $\langle X \rangle$ is encoded **without a fixed point**, as the intersection of all closed supersets.

Two conditions, and an `unsat` means nothing without them. *Non-vacuity*: an unsatisfiable hypothesis makes an implication vacuously true, so `unsat` would prove nothing — each hypothesis is therefore checked satisfiable, and satisfiable with S strictly inside the box rather than filling it. *Encoding fidelity*: the witness formula above must define the same set as $R(X)$, so it is evaluated against an independently derived staircase over random instances, together with a **negative control** that must fail on a deliberately wrong reference — a check that merely restates the formula would confirm the transcription and not the encoding.

Scope of each obligation, stated rather than implied. The obligations for Lemma 4 and Clause B carry the D1 hypothesis, so they range over every subset **satisfying D1**, not every subset simpliciter. Those for Lemma N1* and its reverse carry no such hypothesis. Clause C is discharged in two parts: the **engine** (Lemma 8 — the join-irreducibles regrow X) and the **containment** in `algebra`.

What remains unchecked, and why

Clauses **D, E, F, G, H** and Lemma 7 are **not** machine-checked. They are not of that shape: **D** quantifies over *operators*, **E** and **H** are about five specific named operators, **F** over relabellings, and **Lemma 7** is the cited abstract theorem. Machine-checking them would mean formalizing the five operators themselves, which is a larger undertaking than this document represents.

§11 · Questions raised in review, and their disposition

Every question this work raised about the law is now closed. They are listed with the manner of closing, so a reader can check the closing rather than take it.

1. Do Queyranne–Tardella §4–§6 bear on the law? NO — closed by the paper's own §1, which is transcribed in full. Their introduction states §4 gives *upper and lower bounds on the number of sublattices*, §5 a *corner representation ... for encoding*, and §6 the *sublattice hull membership problem and a polynomial-time algorithm*. The representation theorem this work depends on is §2–§3, and §2–§3 are transcribed complete. Counting, encoding and algorithmics cannot bear on whether the hull is fixed by its 2-fold projections.

2. Is order-convexity the weakest sufficient condition for N1? NO, and the sharp condition is now proved — Lemma N1* and its corollary in §8: $R_B(X) = \langle X \rangle$ iff $R_B(X) \subseteq \text{Box}(X)$. Necessary and sufficient, three lines, and 0 failures in 702,628 exhaustive cases. Convexity is a sufficient condition *for that corollary*.

3. Does the proper-epigraph formulation rescue the non-chain case? NO — closed with a proof, §8 N2a. Strictness only removes cells, so `proper` $\subseteq$ `naive`, while off the chains the failure is *under-generation*. Attainment-strictness cannot repair under-generation.

4. Is `information` `k`-determined for no `k`? CORRECT, and now a theorem for every `d` $\geq$ 3 — the construction $X_d = \{0\} \cup \{e_i + e_d\}$ in §8d, Axis 1. This one was reported open in an earlier draft on the strength of a measurement that turned out to be a size artifact.

5. Machine-checking. DONE for the core, by Z3, and the record says exactly which parts. Lemma 4, Clause B, both parts of Clause C and **both directions of** Lemma N1* are discharged over every subset of the cubic boxes named in §10 — **21 of 21** obligations, `unsat` on each negation, behind two soundness guards. Clauses D–H and Lemma 7 are **not** machine checked, for the reason given in §10: they are not of that logical shape. Each has a written proof and, where applicable, an explicit witness. **The dichotomy is three-way, not two:** a step is checked by a solver, checked by a reader, or **taken from the literature** — and §10's PROVED column reads "cited" for Lemma 7 and "cited + ✓" for Clause B precisely because the lift is imported rather than reproved here. No clause rests on a step that is *unaccounted for*; one clause rests on a step accounted for by citation, and §9 says which citation.

Scope note

Clauses **A–H are the law, and they are complete.** Appendices **§8b** (`analysis`) and **§8c** (`logic`) are **not clauses of the law** — they concern candidate operators outside the five, filed as evidence bearing on which operators belong to the family, and they carry their own status markers. Nothing in A–H depends on either appendix.